Jack Le (Tiep Le)

CTO · Solutions & Technical Architect

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Vietnam · English, Vietnamese

I architect platforms that stay correct under load — and lead the teams that ship them.

18 years from multi-threaded telecom systems to AI-native products. I sit between the business and the engineers: run the solution workshop, draw the C4, write the ADRs, then build and load-test the first version myself. From a 300-engineer delivery division to the CTO seat, across telecom, insurance, banking, healthcare, enterprise SaaS, gaming, Web3 and AgriTech. Today I design AI-native systems: agent memory over MCP, server-authoritative economies, deterministic real-time simulation.

300	96.0%	1.6 ms	4,000	4
engineer division I was Technical Architect for	recall@5 on LongMemEval-S, 500 questions	sim tick p50 with 64 players, 718 units	CCU on 4 shards — one 4-vCPU box that also runs Postgres and Redis	countries on-site with enterprise clients

EXPERIENCE

Principal Engineer & Architect · Independent · AI & platform engineering — 2023 – now

Design and build production platforms end to end, from the first stakeholder conversation to the on-call runbook.

- Architected Xóm Nông Trại: ledger-backed economy, idempotent mutations, outbox, and AI ops agents limited to proposals that humans approve.
- Created Cortex Hub, an open-source MCP platform for agent memory and code intelligence (96.0% recall@5 on LongMemEval-S).
- Built a deterministic Rust/WASM simulation and authoritative multiplayer server, benchmarked to 1.6 ms tick p50 at 64 players.
- Delivered a multi-tenant AgriTech platform and the Kèo Thơm real-time auction marketplace.

TypeScript · Node.js · Rust · PostgreSQL · Redis · GCP · Cloudflare · MCP · LLM

Chief Technology Officer · High-tech agriculture company — 2022 – 2023

Led architecture for a platform running thousands of hectares of production.

- Big-data pipelines processing millions of field data points for real-time monitoring and analytics.
- Crop lifecycle, export-compliance traceability and live operational dashboards on distributed services.
- Kept the system scalable and reliable under continuous data growth.

Microservices · Kubernetes · PostgreSQL · Big data · Real-time

Chief Technology Officer · D2 Solutions — 2021 – 2022

Founded and led the team behind a cross-platform 3D multiplayer game with a Web3 economy.

- Distributed real-time backend built to scale to millions of users with elastic micro-clusters of ~20 players per room.
- Custom low-latency TCP networking layer that removed the dependency on Photon.
- gRPC, Kafka and Redis on Kubernetes (HPA/VPA) with CI/CD and a full observability stack.
- Auditable on-chain / off-chain token economy.

C++ · Unity · gRPC · Kafka · Redis · Kubernetes · AWS · GCP

Technical Architect · Enterprise platforms — 2018 – 2021

Architecture for ERP, logistics and e-commerce platforms serving 100K+ users.

- Microservices on Kubernetes with PostgreSQL and CI/CD on AWS.

Microservices · Kubernetes · PostgreSQL · AWS · CI/CD

Senior Software Engineer · Ansarada — 2016 – 2018

Enterprise SaaS for deal management and virtual data rooms.

- Engineered inside the most disciplined microservices architecture I have worked in: small services, explicit contracts, CI/CD on every change.
- Test-driven development as the norm; kept unit-test coverage above 90% on the code I owned.

- Tuned SQL Server queries and indexes on performance-critical paths — work singled out for praise by the CTO.
- Moved from outsourcing to product engineering: features measured by customer value.

`.NET · Microservices · SQL Server · TDD · CI/CD`

Technical Architect · Senior Engineer · TMA Solutions — 2008 – 2016

Grew from engineer to Technical Architect for the DG1 division (~300 engineers), serving global telecom clients.

- Owned architecture, technical direction and mentoring across the division.
- High-performance multi-threaded C/C++ on Windows, Linux and Solaris; led Windows → Linux and 32 → 64-bit ports down to drivers.
- On-site with customers in the USA (twice, six months each), the UK and Australia (twice).
- Outsourced delivery for banking and healthcare clients (client names under NDA).

`C/C++ · Telecom · Linux · Solaris · Multi-threading`

SELECTED PROJECTS

Xóm Nông Trại · Architect & lead engineer — 2026

Server-authoritative social platform with a real economy

4,000 CCU on 4 shards, one 4-vCPU box shared with Postgres + Redis

5.0 → 2.9 Postgres queries per HTTP request

`TypeScript · Fastify · PostgreSQL 16 · Redis · BullMQ · Three.js — Personal project · private repository`

Cortex Hub · Creator — 2026

One MCP endpoint for AI agent memory and code intelligence

96.0% recall@5 · LongMemEval-S (500 q)

97.8% recall@10 on the same set

`TypeScript · MCP · Qdrant · SQLite · Ollama — github.com/lktiep/cortex-hub`

RA2 Studio · Sole engineer — 2026

Deterministic real-time simulation, native and WebAssembly

1.6 ms tick p50 · 64 players, 718 units

0.8 KB/s bandwidth per player · 0 dropped

`Rust · WebAssembly · tokio · WebGL2 · Cloudflare Workers — Personal project · private repository`

Shrimp Farm Platform · CTO & lead engineer — 2024 – present

Multi-tenant AgriTech SaaS for commercial aquaculture

Dozens farms managed per operator

1000s ponds under monitoring

`Next.js · Supabase · PostgreSQL · TFLite · Gemini — farms.shrimpfarm.vn`

Kèo Thơm · Architect & full-stack engineer — 2025

Real-time auction marketplace with escrowed wallets

`Next.js · Supabase · PostgreSQL · Cloudflare Pages · TypeScript — keothom.bid`

CAPABILITIES

Architecture	Distributed systems · Microservices & event-driven · Real-time & multiplayer · Multi-tenant SaaS · C4 · ADRs · data contracts · Security & threat modelling
AI systems	LLM integration (OpenAI, Gemini, Claude) · OCR & document pipelines · Agents & MCP servers · RAG & vector search · Human-in-the-loop approval · Evaluation & benchmarks
Cloud & platform	Google Cloud · AWS · Kubernetes (HPA/VPA) · Cloudflare Workers · CI/CD & IaC · Observability · OpenTelemetry
Backend & data	Node.js · NestJS · Fastify · Rust · Go · C/C++ · .NET · PostgreSQL · MySQL · Redis · Kafka · BullMQ · gRPC · REST · WebSocket · Ledgers, idempotency, outbox
Leadership & clients	Pre-sales & solution workshops · Exec & client presentations · Teams up to ~300 engineers · Delivery across Product · QA · DevOps · Code review & standards · Mentoring

RECOGNITION

- Winner, CodeProject Visual Studio Extension Contest
- Products reaching 100,000+ users and downloads
- First Vietnamese to solve 300+ Project Euler problems
- Reverse-engineered a 3D online game client at assembly level

Full case studies, architecture notes and benchmarks at jackle.dev